Ch 9.1: Maximum Margin Classifier

Lecture 26 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Mon, Mar 30, 2026

Announcements

Last time:

- Ch 8: Random Forests

This lecture:

- Maximal Margin Classifier
- No jupyter notebook for this class

Announcements:

21	W	3/18	Polynomial & Step Functions	7.1-7.2		
22	F	3/20	Step Functions; Basis functions; Start Splines	7.2-7.4		
23	M	3/23	Regression Splines	7.4		
24	W	3/25	Decision Trees	8.1		Q7
25	F	3/27	Random Forests	8.2.1, 8.2.2	HW #5 Due Sun 3/29	
26	M	3/30	Maximal Margin Classifier	9.1		
27	W	4/1	SVC	9.2		Q8
28	F	4/3	SVM	9.3, 9.4		
29	M	4/6	Single Layer NN	10.1		
30	W	4/8	Multi Layer NN	10.2		Q9
31	F	4/10	CNN	10.3	HW #6 Due Sun 4/12	
32	M	4/13	Unsupervised learning / clustering	12.1, 12.4		Q10
33	W	4/15	Virtual: Project Office Hours			
	F	4/17	***Review***			
	M	4/20	**Midterm #3**			
	W	4/22				
	F	4/24			**Project Due**	

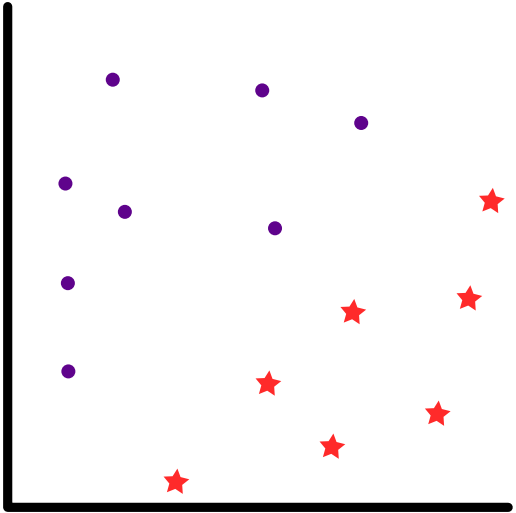
What will you learn today?

- What is hyperplane?
  - How do you mathematically describe this hyperplane?
  - How do you mathematically describe the two sides of the hyperplane?
  - Given the equation of the hyperplane and the coordinates for a point, you should be able to tell which side of the plane the point is on.
- What qualify a hyperplane as a separating hyperplane?
  - You should be able to describe this mathematically using an inequality.
  - You should also be able to determine whether a hyperplane is a separating hyperplane given a graph, or given the equation of the plane and the coordinates and class of a few points.
- How to use a separating hyperplane as a classifier?
- What makes a hyperplane a maximal margin hyperplane?
  - What are its margin and support vectors?
  - Given a graph, you should be able to clearly label the margin and support vectors. You should also be able to infer the size of the margin from reading the graph.
  - You should also be able to describe the optimization problem mathematically.
  - Given the equation of the maximal margin hyperplane and the coordinates of support vectors, you should be able to calculate the size of the margin by hand.

Section 1

Maximal Margin Classifier

The goal



- Point out today is all about classification
- Goal is to find a line in the middle that separates the two classes. Lots of options, but we want the one that stays as far from everyone as possible.
- For today, need to use $y \in \{-1, 1\}$ for this to make sense.

What is a hyperplane?

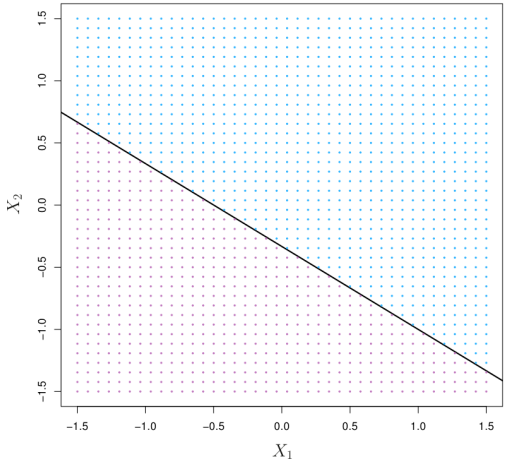
- In p-dimensional space, a hyperplane is a flat affine subspace of dimension $p - 1$
- affine means it doesn't have to pass through the origin
- Draw example for $p = 1, 2, 3$

Mathematical definition of a hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

- Being on the hyper plane means having a point $X = (X_1, X_2, \cdots, X_p)^T$ satisfying this
- Show that when $p = 2$, this is just a line

Hyperplane for $p = 2$

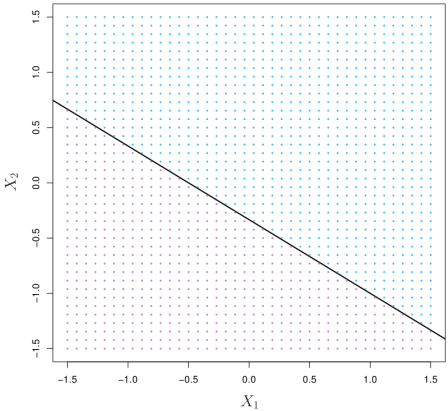


$$1 + 2X_1 + 3X_2 = 0$$

- point of some examples:
- $X_1 = -0.5,\ X_2 = 0$
- $X_1 = 0,\ X_2 = -1$, on lower left with $1 + 2X_1 + 3X_2 < 0$
- $X_1 = 0,\ X_2 = 1$, on upper right with $1 + 2X_1 + 3X_2 > 0$

There are two sides to every hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0 \qquad \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0$$



Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

Observations in one of two classes, $y_i \in \{-1, 1\}$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

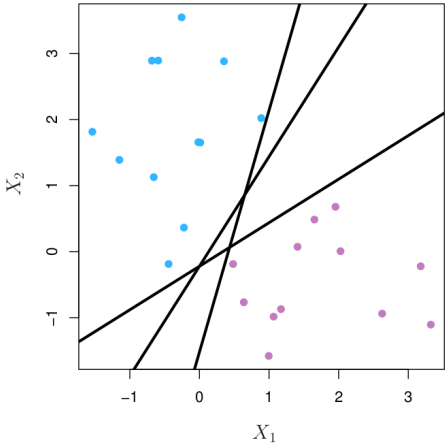
$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Separate out a test observation

$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

Separating Hyperplane

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$



- Strict requirement
- No points can be on the wrong side of the line

Another way to say it

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

For all i:

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

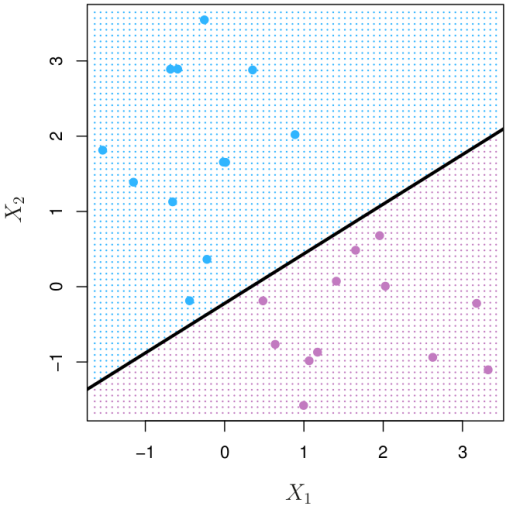
Separating hyperplane becomes a classifier

If you have a separating hyperplane:

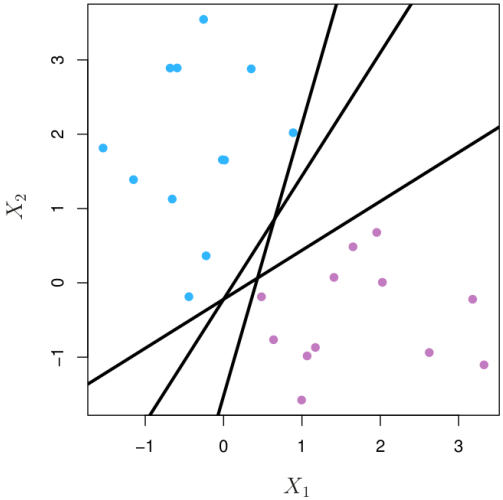
- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$

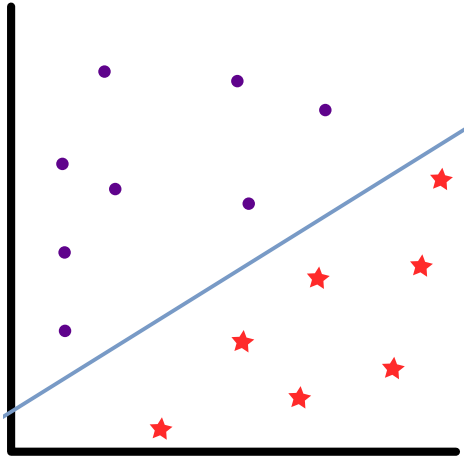


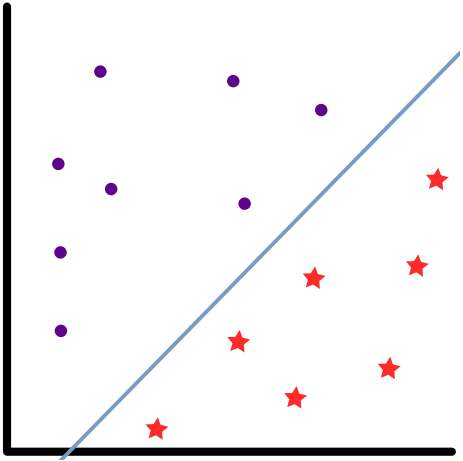
How do we pick?



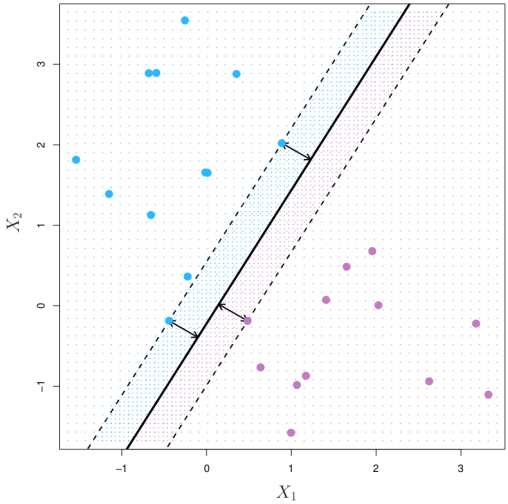
- *There's lots of separating hyperplanes here, so how do we find a "best" one?*
- Compute the perpendicular distance from each training observation to a given separating hyperplane
- Smallest such distance is the *margin*

Distance from an observation to a hyperplane



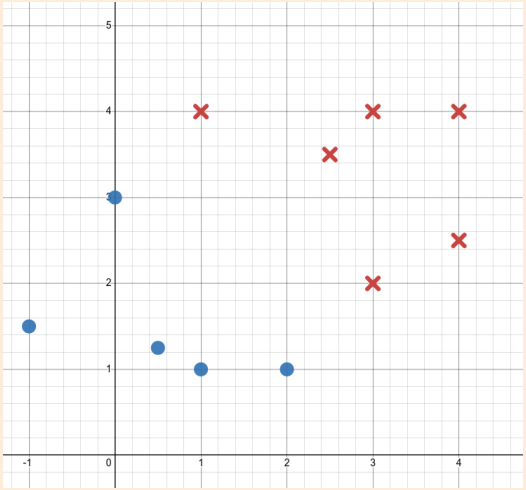


Maximal margin classifier



- *Maximal margin hyperplane* is the hyperplane with the largest minimum distance to the training observations
- Classify using it gives you the *maximal margin classifier*
- Observations that are closest are called *support vectors*
- If points moved slightly then MMH would move
- No effect if any other points moved

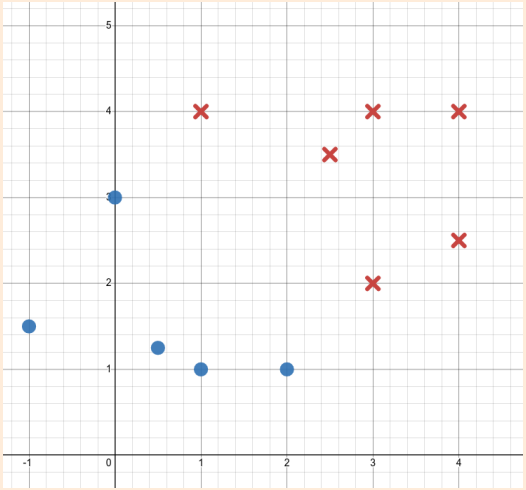
Example



- Sketch the maximal margin hyperplane.
- What is the equation of this line in the form $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$?
- Circle the support vectors. What is their distance from the line?

https://www.desmos.com/calculator/tklbommiwz

Extra work space



Respond to PollEv question.

- $X_1 + X_2 - 4 = 0$
- test points $(0, 0)$, $(4, 4)$, $(2, 2)$
- Points are at distance $\frac{\sqrt{2}}{2}$. Build the isoceles right triangle with d on two sides and 1 for hypotenuse.

https://www.desmos.com/calculator/tklbommiwz

Mathematical Formulation

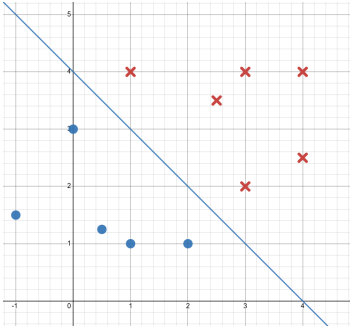
$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

$$\text{subject to} \; \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall \, i = 1, \ldots, n$$

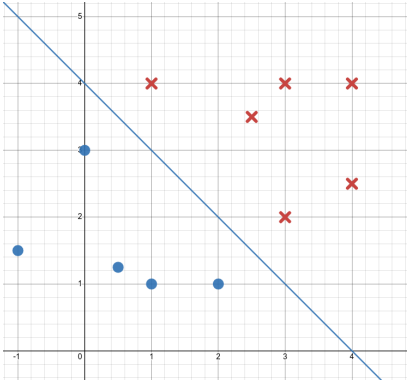
- Last eq forces points to be on the correct side of the hyperplane if $M > 0$
- Making M as big as possible is the maximal margin hyperplane

First constraint



- Had the equation $-4 + X_1 + X_2 = 0$
- Can divde by any number and get the same line.
- $\sum_{i=1}^{p} \beta_i^2 = 1^2 + 1^2 = 2$
- We get the same sloped line if we divide by $\sqrt{(\sum \beta_i)}$,
- Just need to solve for the intercept
- $-\frac{4}{\sqrt{2}} + \frac{1}{\sqrt{2}} X_1 + \frac{1}{\sqrt{2}} X_1 = 0$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2} X_1 + \frac{\sqrt{2}}{2} X_1 = 0$
- Check that $\sum_{i=1}^{p} \beta_i^2$ is now 1
- plug in the support vector $x_i = (1, 4)$, which give $y_i(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2}) = 1/\sqrt{2} =: M$ (the margin)
- Note that $y = 1$ is Red X, $y = -1$ is blue dots below the line

Second constraint

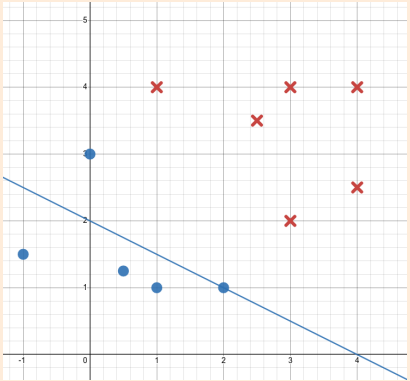


- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_1 = 0$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M$$

- For point $(3, 2)$:
  - $y_i = 1$
  - $1(-2\sqrt{2} + \frac{\sqrt{2}}{2} \cdot 3 + \frac{\sqrt{2}}{2} \cdot 2) = \frac{\sqrt{2}}{2}$
- For point $(0, 3)$:
  - $y_i = -1$
  - $-1(-2\sqrt{2} + \frac{\sqrt{2}}{2} \cdot 0 + \frac{\sqrt{2}}{2} \cdot 3) = \frac{\sqrt{2}}{2}$
- For point $(1, 1)$
  - $y_i = -1$
  - $-1(-2\sqrt{2} + \frac{\sqrt{2}}{2} \cdot 1 + \frac{\sqrt{2}}{2} \cdot 1) = \sqrt{2}$

An example with a bad choice of hyperplane

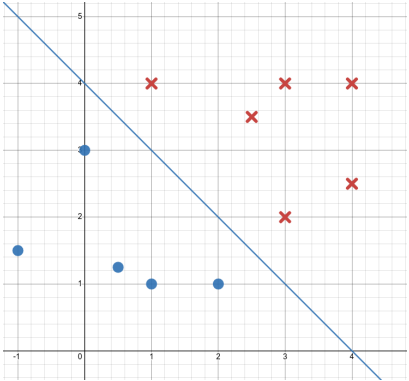


- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-\frac{4}{\sqrt{5}} + \frac{1}{\sqrt{5}}X_1 + \frac{2}{\sqrt{5}}X_1 = 0$

What is $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})$ for the point $x_i = (0, 3)$?

- $-1(-\frac{4}{\sqrt{5}} + \frac{1}{\sqrt{5}} \cdot 0 + \frac{2}{\sqrt{5}} \cdot 3)$
- $-1(-4 + 6)\frac{1}{\sqrt{5}}$
- $\frac{-2}{\sqrt{5}}$
- Note this is negative, so no positive M will make it so this is a valid line

Second constraint extra space



- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_1 = 0$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M$$

- Ensures each on the correct side for a positive M
- When we have unit normal coefficients, the second constraint is distance from the input point to the diagonal.
- Making sure every distance is bigger than M and then make M as big as possible

Mathematical Formulation

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

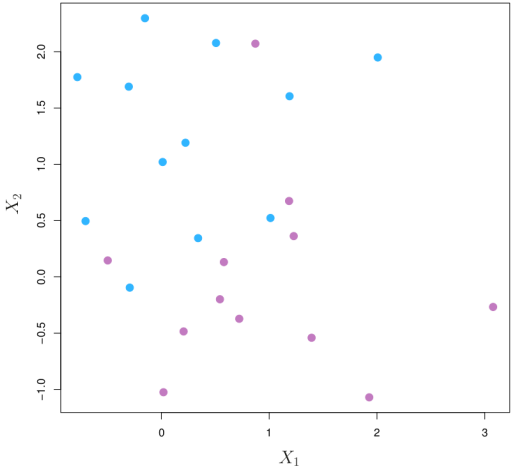
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \quad \forall\, i = 1, \ldots, n$$

- M is making sure every point is at least distance M from the hyperplane
- Because of the y_i multiplied, this also requires that the point is on the correct side of the hyperplane
- Last eq forces points to be on the correct side of the hyperplane if $M > 0$
- Making M as big as possible is the maximal margin hyperplane

Section 2

Issues with Maximal Margin Classifier

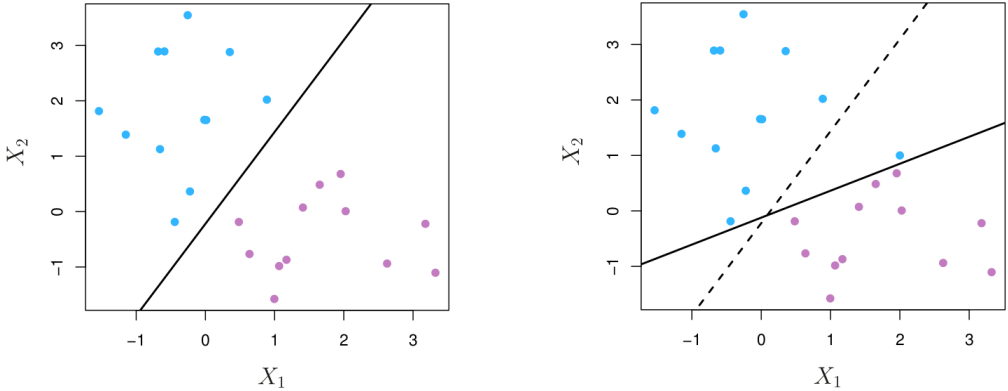
But what if....



- *...there is no separating hyperplane*
- *Means that there is no solution for* $M > 0$

Sensitivity to new points

Even if there is a spearating hyperplane might not be good



- Add a blue observation on the right
- Dramatic shift in maximal margin hyperplane

Next time

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